STUDENT HACKATHON DEVELOPMENT TEAM

# MotionGuard: AI-Powered Road-Safety Intelligence

Turn unannotated traffic video into structured, evidence-backed safety dossiers for faster, more explainable investigations.

“From video pixels to explainable safety intelligence.”



THE EVIDENCE GAP

# Millions of Hours of Video, Zero Automated Evidence

Road-safety teams still face a CCTV blindspot: footage exists, but the evidence is slow to find, noisy to interpret, and difficult to assemble.

- **Manual scrutiny:** Authorities and fleet operators review footage frame by frame after incidents, spending hours identifying the vehicles involved.
- **Bounding boxes are not intelligence:** Object detectors show where an entity is—not how it moved or why an interaction became dangerous.
- **Alert fatigue:** Frame-level proximity heuristics can produce dozens of duplicate warnings for one near-miss.
- **Evidence disconnect:** Incident reports require timestamps, images, and vehicle IDs to be stitched together across disconnected tools.



THE MOTIONGUARD PIPELINE

# MotionGuard Turns Raw Footage into an Investigation Dossier

A unified pipeline converts traffic video into traceable interactions, visual proof, and an exportable incident record.

## 01

### Ingest and detect

YOLO11n identifies vehicles, pedestrians, trucks, and buses.

## 02

### Track through time

ByteTrack preserves identities through occlusions and dynamic road scenes.

## 03

### Measure motion

Trails, trajectories, and velocity vectors quantify movement in px/s.

## 04

### Filter interactions

Closing interactions stay; parallel driving and redundant alerts drop out.

## 05

### Export evidence

A tamper-evident PDF dossier carries proof and investigation context.

ENGINEERING STACK

# Engineered for Precision, Speed, and Traceability

MotionGuard combines production-grade computer vision with transparent Python-based investigation tooling.

## Vision and tracking

Ultralytics YOLO11n delivers low-latency detection at approximately 15 ms inference, paired with ByteTrack and Kalman-filter motion priors.

## Trajectory and interaction engine

Image-space displacement tracking, IoU overlap suppression, and relative-velocity dot products turn detections into interpretable motion evidence.

## Aggregation and reporting

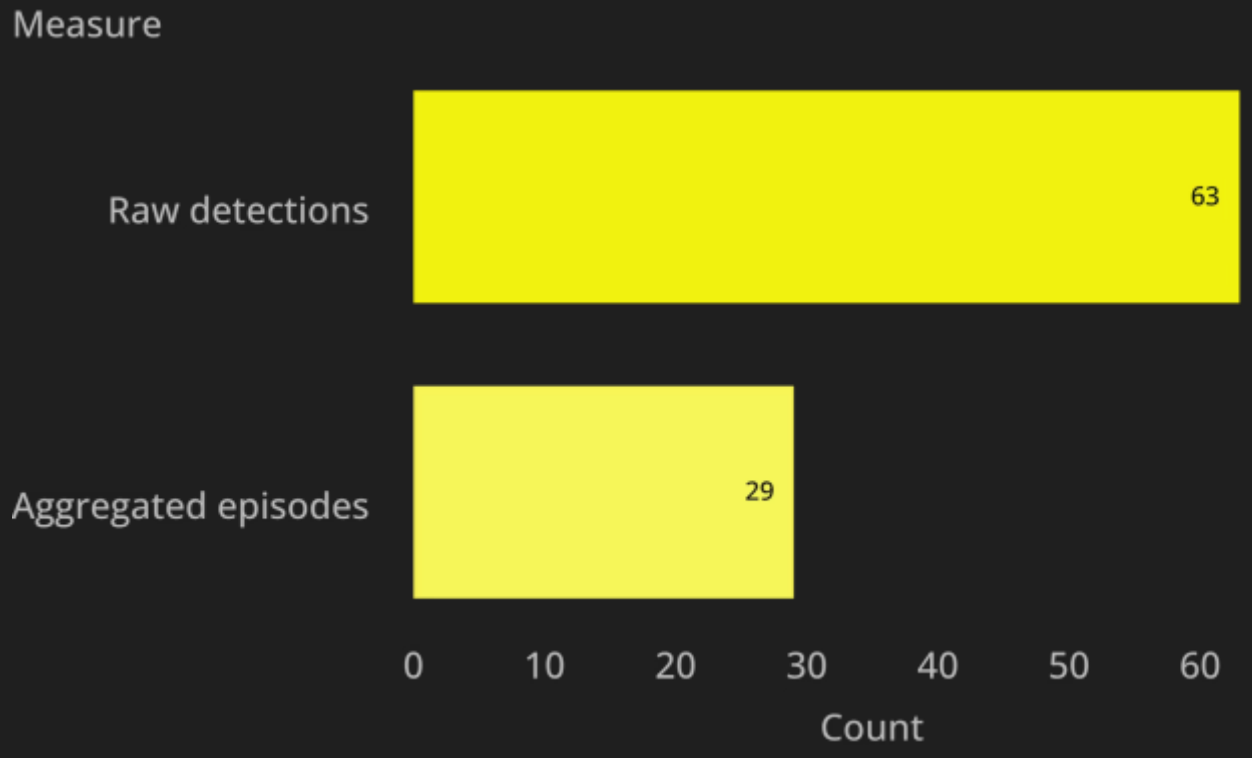
Temporal clustering groups frame-level detections into interaction episodes; Streamlit, OpenCV, and ReportLab power the command center and forensic PDF engine.



BENCHMARK · 716-FRAME CLIP

# 54.0% Less Alert Noise Without Losing Visual Evidence

Temporal event aggregation turns a chaotic stream of frame detections into coherent investigation episodes.




**54.0%**

NOISE REDUCTION

At 30 FPS, a 2-second close approach can create more than 60 raw alerts. MotionGuard groups consecutive timestamps into one episode, such as **PI-0004**, and promotes the closest approach as the primary evidence frame.

Underlying visual evidence remains available.

INTEGRITY BY DESIGN

# No Black Boxes. No False Claims.

MotionGuard helps investigators trust the evidence by making every limitation and rationale explicit.



## Honest kinematics

Monocular uncalibrated cameras distort perspective. The prototype reports image-space velocity in px/s instead of fabricating km/h values.



## Candidate interactions

Proximity flags a situation for human review. It does not claim a collision or autonomous legal certainty.



## Explainable rationale

Threat scores from LOW to CRITICAL surface their mathematical contributing factors, while simulated flags are disclosed as demonstrations.



THE COMMAND CENTER

# From Video to Courtroom Dossier in 60 Seconds

Five tabs connect analysis, evidence, explanation, and export.

01

Analyze video with ByteTrack IDs and historical trajectory trails.

1

02

Open PI-0004: minimum separation 10.12 px; closing speed 54.68 px/s.

2

03

Review the generated snapshot with metadata and highlighted vehicles.

3

04

Test what-if overspeed scenarios and inspect rule-based risk scores.

4

05

Dispatch live Telegram/SMS alert to field units and export courtroom PDF dossier in 60s.

5

ROADMAP

# Scale Explainable Safety Intelligence to the Roadside

MotionGuard's prototype establishes the evidence loop. The roadmap extends it toward infrastructure-grade traffic safety.

## 01

Phase 1 · Current

Local prototype, trajectory analysis, event aggregation, and PDF reporting.

## 02

Phase 2

Camera homography self-calibration for metric ground-plane speed in km/h.

## 03

Phase 3

Edge deployment on roadside CCTV units with real-time alerting.

## 04

Phase 4

Multi-camera handoff for continuous vehicle tracking across complex intersections.

**Targets:** reduce post-incident review time by 80% and maintain a 100% link between findings and source frames.



PROJECT REPOSITORY

# Explore MotionGuard

Browse the source code, documentation, and project details on GitHub.

Open the GitHub project

github.com/vishalbhardwaj-source/MotionGuard